Report proofreading and review

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Questions Question 1 – Conceptual/E-R modelling: design a conceptual/Entity-Relationship (E-R) model that addresses the database context. Your mo…

PASTED

Please check spelling, punctuation and grammar, to format the report and check its structure, to check that every question in the brief had been answered, and to check that each answer had enough
detail

Compiling the review's spelling and layout fixes

Review of your ST207 Assignment 1 report

The report is well written, internally consistent, and clearly structured. I checked the figures quoted in your text against the screenshots (customer 135, Zenith laptop 34.3%, 59.5% Champions, the Nov 2025 peak, the 840-order base implied by the support percentages) and they match. The main gaps are missing evidence for Q1 and Q2, a few constraint decisions that are hard to defend, and some wording and formatting inconsistencies.

One limitation: pages 1-4 reached me as text only, so I couldn't see any diagrams. If your E-R and relational diagrams are on those pages, ignore the diagram points below.

1. Coverage against the brief

Question	Status	Gap
Q1 E-R model	Strong explanation	No visible reference to the diagram or to Crow's foot notation. The brief also asks you to verify consistency with the database context before Q2, and there is no explicit statement of that check. Design decisions are justified, but not tied to named use cases.
Q2 Relational model	Strong	Data types and nullability per attribute aren't shown in the text, so a schema listing or table would help. The constraints are thin (see section 2).
Q3 Creation	Good	FK handling and view-vs-table decisions are covered. Add a pointer to the DDL file or a short excerpt.
Q4 Data loading	Very good	Add a products-per-category distribution, since the brief asks about distribution across categories.
Q5 SQL	Good	Each use case needs a one-line statement (who uses it, and why). The brief lists UDFs among the constructs, and you don't explain their absence.
Q6 Triggers	Good	Pass and fail demos are present for both sets. The trigger code isn't shown, and a small table listing each trigger would help.

2. Substantive issues worth fixing

1. **Constraints.** You state that UNIQUE is used only once, and that CHECKs and composite keys aren't used because "the data generator follows them." Markers will likely see that as weak, because the database should protect itself regardless of the loader. Your own validation queries check for duplicate emails, SKUs and payment references, one review per customer per product, ratings 1 to 5, positive prices and no variant twice in a cart or order. All of these can be declared in SQLite:
   - UNIQUE on `Customer.email`, `ProductVariant.sku` and `Payment.transaction_ref`
   - composite PRIMARY KEY on `CartItem(cart_id, variant_id)` and `OrderItem(order_id, variant_id)`
   - UNIQUE on `Review(customer_id, product_id)`
   - CHECK on rating, price, quantity, stock and the status and method lists

   If you add them, re-run your scripts and validation queries and update the Q2/Q3 text. If you can't, replace "the generator follows them" with a proper justification.
2. **Missing diagrams and schema listing.** Q1 requires Crow's foot notation. The text uses 0..N and 0..1 style cardinalities, so say explicitly which figure shows the notation. Also add a compact schema listing, for example `OrderItem(order_id PK/FK, variant_id PK/FK, quantity, unit_price_at_purchase)`.
3. **Consistency check.** Add a short paragraph before Q2 mapping the database-context requirements to your entities. This is the "verify consistency" instruction in Q1.

4. **Data dictionary.** Your AI statement says Claude helped build one, but the report never mentions it. Either say where it is or remove the claim.
5. **Numbers to reconcile.**
   - "49 customers with no orders" (Q4) versus "51 customers have no completed order" and "Never bought: 51" (Q5). Presumably 2 customers have only cancelled orders, so say so.
   - "Completed" clashes with the Payment status "Completed", so use "no non-cancelled order".
   - Query 5.4 says it shows the last twelve complete months, but the table starts in Oct 2025, while your data reaches at least 26 Oct 2026. Check the window against your data end date.
   - Demo 2C: `shipped_at` is 03:27:24 but the log entry is 03:35:01. If one trigger sets both in one statement they should match, so check you haven't mixed two runs.
6. **Use case labelling.** Use Case 1 is titled "customer details and orders" but includes Query 5.2, a marketing email list. Retitle the use case, or move 5.2 into marketing.
7. **UDFs.** Add one sentence, such as "SQLite has no CREATE FUNCTION, so user-defined functions weren't used; views serve the reuse role."
8. **Minor modelling point.** "Disjoint" is trivially true with only one subtype (CardPayment), so you could say so. Also, `city` depends on `postcode` in Address, which strictly breaks 3NF. Either note this as a deliberate simplification or soften the "third normal form" claim.

3. Spelling, punctuation and grammar

Your UK spelling is consistent and I found no outright typos. Suggested edits:

- **"Product records identify displayed items through product_id."** → "Each Product is identified by product_id and represents an item as shown in the catalogue."
- **"This is kept by the data and checked with a validation query."** → "This is guaranteed by the loaded data and checked with a validation query."
- **"e-mail"** (Query 5.1 caption) → "email", to match the rest of the report.
- **"The report shows that…"** (twice in Q5) → "The output shows that…". "The report" is ambiguous here, since it also means the whole document.
- `RAISE(FAIL, 'ERROR: ...')` begins with a closing curly quote. Use straight quotes in code font: `RAISE(FAIL, 'ERROR: ...')`.
- **"…added to inventory_log; in contrast:"** → "The second test, requesting 5 units, was rejected:". The same applies to the later "in contrast:" → "In contrast, a legal transition succeeded:".
- **"which it works out from the order lines, the shipping fee and the completed payments"** is ambiguous. → "The order total is calculated from the order lines and the shipping fee, then compared with the sum of completed payments."
- **Query 5.3:** "SUM() OVER (PARTITION BY category) gives each product's share" → "gives the category total, from which each product's share is calculated."
- **Query 5.5:** add one clause defining support (share of all orders containing both products) and confidence (share of orders with A that also contain B).

- **Voice:** you alternate between "I loaded…" and passive constructions. Either is fine, but pick one.

4. Formatting and structure

- **Headings** don't follow the brief's numbering. Suggested: "Question 1 – Conceptual (E-R) model", "Question 2 – Relational model", "Question 3 – Database creation", "Question 4 – Data loading" (currently "Data Sources:"), then Questions 5 and 6.
- **Trailing colons** in headings ("Question 5, Use Case 1: Customer details and orders:", "Database Creation:", "Customer:") should go, or be applied consistently.
- **"Question 6i)"** → "Question 6, Part (i)".
- **Q5 layout:** the explanations of 5.1 and 5.2 are interleaved across paragraphs. Use a repeating pattern per query: purpose, how it works, constructs used, finding. Keep the same pattern for 5.3-5.6.
- **Figures** should be numbered ("Figure 1…") and referred to in the text.
- **Screenshots** from DB Browser are small. Crop tighter or enlarge them so the text is readable at 100%.
- **Query and trigger code** isn't in the report. Say which attached file contains each, or add a short appendix.
- **Referencing:**
  - There are no in-text citations. Cite (ST207, 2026) where you mention the seminar pattern and `emp_log`, and (SQLite Tutorial, n.d.) near the window functions.
  - Reference 2 lacks an access date, and reference 3 lacks a URL and a model description.
  - The AI use is described twice, in reference 3 and in the statement. Shorten reference 3 to a plain citation and keep the detail in the statement.
- If there is a word or page limit, check it after adding the above.

If you'd like, I can draft the consistency-check paragraph, the trigger summary table, or a cleaned version of the prose sections.